

L 14049 Applied Behavioural Economics

Lecture 2: Heuristics and biases in judgement and decision making

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L 14049 Applied Behavioural Economics

Lecture 2 and 3: Heuristics and Biases in Judgement and Decision making

Recommended readings:

Bazerman, Max H. and Don A. Moore, 2009. Judgement in managerial decision making, Wiley (7th edition), Chapter 2

Tversky, Amos and Daniel Kahneman, 1974. Judgment under uncertainty: Heuristics and biases, Science, pp. 1124-1131

Tversky, Amos and Daniel Kahneman, 1973. Availability: A heuristic for judging frequency and probability. Cognitive Psychology, pp. 207-232

L 14049 Applied Behavioural Economics

Lecture 2 and 3: Heuristics and Biases in Judgement and Decision making

Further readings:

Casscells, Ward, Arno Schoenberger and Thomas Graboys. 1978. Interpretation by physicians of clinical laboratory results. New England Journal of Medicine 299 (18). pp. 999-1000

Clotfelter, Charles, and Philip Cook. 1993. The 'Gambler's Fallacy' in Lottery Play. Management Science, 39(12), pp. 1521-25.

Epley, Nicholas and Thomas Gilovich, 2006. The Anchoring-and-adjustment Heuristic. Psychological Science, 17 (4), pp. 311-318

Guryan, Jonathan and Melissa S. Kearney, 2008. Gambling at Lucky Stores: Empirical Evidence from State Lottery Sales, American Economic Review, pp. 458-473

Northcraft, Gregory and Margaret Neale, 1987. Experts, Amateurs, and Real Estate: An Anchoring-and-Adjustment Perspective on Property Pricing Decisions. Organizational Behaviour and Human Decision Processes, 39, pp 84-97

Wason, P.C., 1968. Reasoning about a rule. Quarterly Journal of Experimental Psychology, pp. 273-281

Judgements and decision making

Introduction to judgemental heuristics

Heuristics (rules of thumb) are cognitive tools that are used to simplify decision making

Problem: heuristics are often useful but they can also lead to systematic errors

Judgements and decision making

Psychology distinguishes two models of thinking
(Stanovich and West, 2000)

System 1 thinking: quick and intuitive thinking, effortless, emotional, associative, difficult to control or modify

System 2 thinking: conscious reasoning, slow, effortful, used for complex demanding tasks

Judgements and decision making

Psychology distinguishes two models of thinking
(Stanovich and West, 2000)

Problem: System 2 is lazy

People avoid cognitive effort and often trust a plausible judgment that quickly comes to mind

Judgements and decision making

Example: Problem 1

A bat and a ball cost £1.10. A bat costs £1 more than a ball. How much does the ball cost?

Intuitive answer: 10 pence **Correct answer:** 5 pence

ball	£ 0.10	ball	£ 0.05
<u>bat</u>	<u>£ 1.10</u>	<u>bat</u>	<u>£ 1.05</u>
set	£ 1.20	set	£ 1.10

Your answers:

32 % said 10 pence	(last year 21%)
64 % said 5 pence – well done!	(last year 78%)
4% other answers	(last year 1%)

Judgements and decision making

Introduction to judgemental heuristics and biases

1. Availability
 - Ease of recall
 - Retrievability
2. Representativeness
 - Insensitivity to base rates and sample size
 - Conjunction fallacy
 - Misconception of chance
 - Regression to the mean
3. Confirmation heuristic
 - Confirmation trap
 - Anchoring and insufficient adjustment
 - Conjunctive and disjunctive event bias
 - Overconfidence
 - Hindsight and the curse of knowledge

Heuristics and Biases

1. Availability

- People assess the frequency of a class or the probability of an event by the ease with which instances or occurrences can be brought to mind
- Often a good clue for the judgment of frequencies: In general, frequent events are easier to recall than infrequent ones
- But not always....

Heuristics and Biases

Availability - Bias: Ease of recall based on vividness

Problem 2:

The following 10 corporations were ranked by Fortune magazine to be among the 500 largest worldwide according to revenue for 2014. Which group of five organizations listed has the larger total revenue?

Group A: China Electronics (382), McDonald's (430), Accenture (404), Danone (429), L'Oréal (402)

Group B: Petrobras (28), McKesson (29), Enel (56), Verizon Communications (42), Munich Re Group (93)

(Adapted from Bazermann, 1994, actual ranks in parentheses)

Your answer: Group A 46 % (last year 52%)

Correct answer: Group B 54 %

Group B contains firms that are less well known consumers (less vivid).

Heuristics and Biases

Availability - Bias: Ease of recall based on vividness

Problem 3:

Which of the following causes more death in the UK each year?

- Stomach cancer (in 2008: 5178)
- Motor vehicle accidents (in 2008: 2538)

Your answer: **64 % for b.**

Typical answer: b.
Motor vehicle accidents more prevalent in the media.

Russo and Shoemaker (1989) counted in two typical newspapers over a period of one year 137 reports on death caused motor vehicle accidents, but only one by stomach cancer

Heuristics and Biases

Availability - Bias: Retrievability

Problem 4: Version A

In four pages of a novel (about 2,000 words), how many words would you expect to find that have the form _ _ _ _ _ n _ (seven-letter words that have the letter n in the sixth position)?

Problem 4: Version B

Problem 4: Version B
In four pages of a novel (about 2,000 words), how many words would you expect to find that have the form _____ing (seven-letter words that end with -ing)?

Your answers: Version A: **avg. 13.2** (previous year: 12.03)
 Version B: **avg. 29.7** (previous year: 17.00)

Heuristics and Biases

Availability - Bias: Retrievability

Problem 4: Version A

In four pages of a novel (about 2,000 words), how many words would you expect to find that have the form _ _ _ _ _ n _ (seven-letter words that have the letter n in the sixth position)?

Problem 4: Version B

In four pages of a novel (about 2,000 words), how many words would you expect to find that have the form _ _ _ _ _ ing (seven-letter words that end with -ing)?

Most people estimate higher numbers for version B: -ing words are more easily retrievable

All seven letter words ending with "ing" also have an "n" as their sixth letter

Thus, also **conjunction fallacy**: $p(A \& B)$ cannot be greater than $p(A)$ or $p(B)$

Heuristics and Biases

Availability and combinations

Problem 5:

In the drawing below there are ten stations along a route between Start and Finish. Consider a bus that travels, stopping at exactly **2 (or 8)** stations.

Estimate the number of different patterns of **2 (or 8)** stops that the bus can make?

Start

--	--	--	--	--	--	--	--	--	--

 Finish

Average estimate for 2 stations: **53** (previous year 96.2)

Average estimate for 8 stations: **25.4** (previous year 57.4)

Correct: The number of patterns of r stops is given by the binomial coefficient

$$\binom{10}{r} = \frac{10!}{r!(10-r)!}$$

For $r = 2$ and 8 the correct solution is 45:

$$\frac{10!}{2!(10-2)!} = \frac{10!}{2!(8)!} = \frac{10!}{8!(10-8)!} = \frac{10!}{8!(2)!}$$

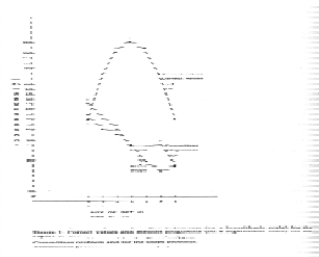
Heuristics and Biases

Availability and combinations

Problem 5: Original experiment by Tversky and Kahnemann (1973)

- People perceive to have "more degrees of freedom in constructing patterns of two stops."

- Patterns of two stops are more available: "more such patterns are seen at first glance, they are more distinctive, and they are easier to visualize".



Heuristics and Biases

Availability heuristic - Summary

- Judgement can be biased by information that is easy to recall, vivid or recent
- This is often appropriate and yields adequate decisions
- But it also can cause systematic errors

Heuristics and Biases

Representativeness

Judgements follow stereotypes:

The probability that an object or event A belongs to a class B, is evaluated by the degree to which A is representative of B (or by the degree to which A resembles B).

Again, this heuristic can lead to serious errors...

Heuristics and Biases

Representativeness –

Bias: Insensitivity to base rates (prior probabilities)

Example: Kahneman and Tversky (1974)

Subjects were shown brief personality descriptions of several individuals, allegedly sampled at random from a group of 100 professionals - all engineers or lawyers. The subjects were asked to assess, for each description, the probability that it belonged to an engineer rather than a lawyer.

Two experimental conditions:

1. Subjects were told the group consisted of **70** engineers and **30** lawyers
2. Subjects were told the group consisted of **30** engineers and **70** lawyers

Heuristics and Biases

Representativeness – Bias: Insensitivity to base rates (prior probabilities)

Example: Kahneman and Tversky (1974)

Subjects were shown brief personality descriptions of several individuals, allegedly sampled at random from a group of 100 professionals - all engineers or lawyers. The subjects were asked to assess, for each description, the probability that it belonged to an engineer rather than a lawyer.

Results: very similar probability judgements in both conditions.

Subjects ignore prior probabilities and judge the likelihood of a description being an engineer rather than a lawyer by the degree to which it was representative of the two stereotypes

Heuristics and Biases

Representativeness – Bias: Insensitivity to base rates (prior probabilities)

Subject also ignore base rates if there is **no** relevant information related to the profession (engineer or lawyer).

Example:

Dick is a 30 year old man. He is married with no children. A man of high ability and high motivation, he promises to be quite successful in his field. He is well liked by his colleagues.

Subject's judgement of him being an engineer:
.5 in both groups (irrespective of base rate)

Heuristics and Biases

Representativeness – Bias: Insensitivity to base rates

Problem 6: Harvard Medical School Test
(Casscells, Schoenberger and Graboys, 1978)

60 students and staff at Harvard Medical School were asked:
A particular disease has a prevalence of 1/1000 people. A test to detect this disease has a false positive rate of 5%. Assume that the test diagnoses correctly every person who has the disease. What is the chance that a randomly selected person found to have a positive result actually has the disease?

Results:

Almost half gave the response 95%.
The average answer was 56%.
Only 18% gave the correct answer.

Heuristics and Biases

Representativeness – Bias: Insensitivity to base rates

Problem 6: Harvard Medical School Test
(Casscells, Schoenberger and Graboys, 1978)

Your Results: - 55% gave response > or = 95%
- 40% gave the response 95% +/-1 (last year 30%)
- 18% correct answers (+/- 1%)

The correct answer is **2 %!**

Bayes' rule:

$$\Pr(P|+) = \frac{\Pr(+|P)\Pr(P)}{\Pr(+|P)\Pr(P) + \Pr(+|A)\Pr(A)} \approx \frac{0.001}{0.001 + 0.05 \cdot 0.999} = 0.02$$

P = disease is present, A = disease is absent, + = positive test result

Testing 1000 people yields 49.95 false positive results!
Only 1 out of 50.95 people with positive result has the disease.

Heuristics and Biases

Representativeness – Bias: Insensitivity to sample size

Problem 7 (Kahneman and Tversky, 1974):

There are two hospitals of different sizes in the same town. In the large hospital, 45 babies are born each day, whereas only 15 are born in the smaller hospital. 50% of all babies are boys, but on some days the percentage will be higher and on other days, it will be lower. Which hospital would you expect to record more days per year, when over 60% of the babies born were boys?

your answers:

- ☐ The larger hospital 14% (22%)
- ☐ The smaller hospital 38% (22%)
- ☐ About the same **48%** **(56 %)**

Results in parentheses (K&T 1974)

Heuristics and Biases

Representativeness – Bias: Insensitivity to sample size

Problem 7 (Kahneman and Tversky, 1974):

There are two hospitals of different sizes in the same town. In the large hospital, 45 babies are born each day, whereas only 15 are born in the smaller hospital. 50% of all babies are boys, but on some days the percentage will be higher and on other days, it will be lower. Which hospital would you expect to record more days per year, when over 60% of the babies born were boys?

Typical answer:

- ☐ About the same (that is within 5 % of each other) (56%)

Correct answer:

- ☐ The smaller hospital (22%)

Events seem equally representative

A quick task for you

Buying a company

Your task is to submit a bid for a company.
The current owner of the company knows the exact value of the company to them. When you decide what to bid, you will not know what this value is.

However, you know three things:

1. The **value of the company to the current owner** is equally likely to be any number of between 0 and 100.
2. You will acquire the company if your bid is greater than or equal to the **value of the company to the current owner**.
3. The **value of the company to you** is $1\frac{1}{2}$ times the value of the company to its current owner.

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Lecture 3: Heuristics and biases in judgement and decision making

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Last lecture: Heuristics and biases:

1. Availability
 - Ease of recall
 - Retrievability
2. Representativeness
 - Insensitivity to base rates and sample size
 - Misconception of chance
 - Regression to the mean
 - Conjunction fallacy

Heuristics and Biases

Representativeness – Bias: Misconception of chance

“Chance is commonly viewed as a self-correcting process in which a deviation in one direction induces a deviation in the opposite direction to restore the equilibrium. In fact, deviations are not “corrected” as a chance process unfolds, they are merely diluted.”

Kahnemann und Tversky (1974, p.1125)

Heuristics and Biases

Representativeness – Bias: Misconception of chance

Example (Kahneman and Tversky, 1974):

A coin is tossed six times. Which of the following sequences is most, second most/least likely.

- ☐ H T H T T H
- ☐ H H H H T H
- ☐ H H H T T T

Typical answer: H T H T T H is more likely than H H H T T T

H H H T T T is more likely than H H H H T H

Correct answer: all three are equally likely

Heuristics and Biases

Representativeness – Bias: Misconception of chance

Related phenomena:

- gamblers fallacy (negative serial correlation)
- hot-hand phenomenon (positive serial correlation)

Study by Clotfelter and Cook (1993):

lottery betting on a given number drops off sharply in the days after that number won (negative serial correlation)

Study by Guryan and Kearney (2008): after a sale of a (single) large-prize winning ticket the “lucky” store’s ticket sales increase significantly (positive serial correlation)

Heuristics and Biases

Representativeness – Bias: Misconception of regression

Regression to the mean: if an extreme variable is observed in a first measurement it will tend to be closer to the average in a second measurement.

Problem:

1. In many instances people do not expect regression (tall parents will have tall children)
2. People often invent causal explanation and regression effect is mistaken for a treatment effect

Heuristics and Biases

Representativeness – Bias: Misconception of regression

Regression to the mean: if an extreme variable is observed in a first measurement it will tend to be closer to the average in a second measurement.

Classic examples:

- Overestimation of the effects of punishments and underestimation of the effects of rewards on future performance (a good performance is likely to be followed by a worse one and vice versa)
- Overestimation of the effectiveness of medical treatment (e.g. depressed students treated with chocolate improve significantly over a three-month period)

Heuristics and Biases

Summary: Representativeness Heuristic

We have examined five biases associated with representativeness

- Insensitivity to base rates
- Insensitivity to sample size
- Conjunction fallacy (see problem 4 or Bazerman and Moore problem 8 "Linda Problem")
- Misconception of chance
- Regression to the mean

Again, judgement according to representativeness can be useful, but it also can lead to systematic errors.

Another rule of thumb: Confirmation heuristic

Heuristics and Biases

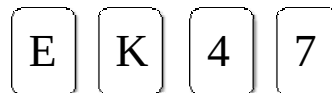
Confirmation heuristic – Bias: Confirmation trap

Tendency to seek evidence that would confirm a hypothesis and to ignore negative evidence

Problem 8: Wason Selection task (1966, 1968)

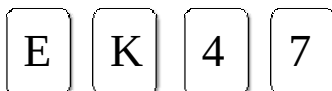
Which of the cards do you need to turn over to obtain conclusive evidence of the following rule:

A card with a vowel on it will have an even number on the other side.



Heuristics and Biases

Confirmation heuristic – Bias: Confirmation trap



Correct answer:

E (positive evidence) and 7 (negative evidence)

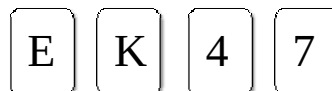
Typical answer: E and 4 (46%)

More than a third choose E only (do not seek negative evidence)

Related phenomenon: Focalism
people focus only on a subset of available information (overweigh attended information, underweigh unattended information)

Heuristics and Biases

Confirmation heuristic – Bias: Confirmation trap



Your answers:

correct answer (E and 7): only 2 out of 22 (2.6%)

Most popular:

E and 4	36%
only E	14%

Heuristics and Biases

Confirmation heuristic

Bias: Anchoring and Adjustment

- A reference point (anchor) significantly influences value estimates
- Anchor can be arbitrarily chosen
- Value estimates are insufficiently adjusted away from the anchor

Heuristics and Biases

Confirmation heuristic- Bias: Anchoring and Adjustment

Problem 9:

Was Ghandi more or less than **42 (92)** years old when he died?

- ☐ more (chosen by 100% with anchor 42)
- ☐ less (chosen by 63% with anchor 92)

How old was Ghandi when he died?

Correct answer: 78 (1869-1948)

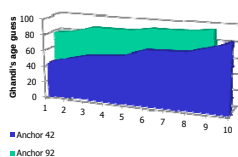
Heuristics and Biases

Confirmation heuristic- Bias: Anchoring and Adjustment

Problem 9: January 2015

Was Ghandi more or less than **42 (92)** years old when he died?

Distribution of guesses



Answer:

Version 1 (anchor 42): average: 64.2 (range: 42-87)

Version 2 (anchor 92): average: 87.4 (range: 77-95)

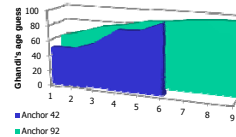
Heuristics and Biases

Confirmation heuristic- Bias: Anchoring and Adjustment

Problem 9: Results January 2014

Was Ghandi more or less than **42 (92)** years old when he died?

Distribution of guesses



Answer:

Version 1 (anchor 42): average: 68 (range: 50-92)

Version 2 (anchor 92): average: 82 (range: 60-95)

Heuristics and Biases

Confirmation heuristic - Bias: Anchoring and Adjustment

Problem 10:

Please write down the last **two/three** digits of your student ID number ____

A survey of more than 80 universities in the UK has found that more than 17,000 incidents of academic misconduct among students were recorded by universities in the 2009-10 academic year.

Do you think the number of cases of academic misconduct at the University of Nottingham is higher or lower than the number you have noted above (last **two/three** digits of your student ID)?

- ☐ lower
- ☐ higher

What is your best estimate of the correct number of academic misconduct among students in the 2009-10 academic year at the University of Nottingham.

Heuristics and Biases

Confirmation heuristic - Bias: Anchoring and Adjustment

Problem 10: Your answers:

Version 1:
two digits

Version 2:
three digits

30	2
37	40
50	100
62	200
64	200
74	300
94	500
100	528
200	910
346	
373	

Average: **130** **309**

Correct answer: 196

(Source: <http://www.telegraph.co.uk/education/educationnews/8363783/University>)

Heuristics and Biases

Confirmation heuristic - Bias: Anchoring and Adjustment

- Anchoring effects are very robust (e.g. survive with monetary incentives and outside the lab)

Example: Anchoring and Adjustment in Property pricing
Northcraft and Neale (1987)

Experiments

- Task: Estimate value of a property
- Subjects: real estate agents (experts), students
- Subjects had a tour through the property and received a 10 page information packet, which was the same with the exception of the appraised value
- Four conditions: 4% (or 12%) above or below the appraised value

Heuristics and Biases

Confirmation heuristic - Bias: Anchoring and Adjustment

Example: Anchoring and Adjustment in Property pricing
Northcraft and Neale (1987)

Results:

Values were significantly biased towards the appraised values (anchors).

Anchoring index:

41% for estate agents

48% for students

Anchoring index defined: (difference between the estimates)/(difference between anchors)

- anchoring index 100% if subjects adopt the anchor as their estimate
- anchoring index is 0% if subjects ignore anchor

Interesting: students admitted that anchor influenced their estimate, estate agents proudly denied that it had any influence

Heuristics and Biases

Confirmation heuristic - Bias: Anchoring and Adjustment

Anchoring index defined: (difference between the estimates)/(difference between anchors)

- anchoring index 100% if subjects adopt the anchor as their estimate
- anchoring index is 0% if subjects ignore anchor

Example: Gandhi task:

Difference estimates: $97.4 - 64.2 = 23.2$

Difference anchors: $92 - 42 = 50$

Anchoring index: $23.2/50 = 46.4\%$

Heuristics and Biases

Confirmation heuristic - Bias: Anchoring and Adjustment

Anchoring effects can affect decisions in a wide variety of situations:

- advertising (offers in a supermarket)
- negotiations (first mover can set the anchor)
- bidding in auctions (estimate)
- surveys...

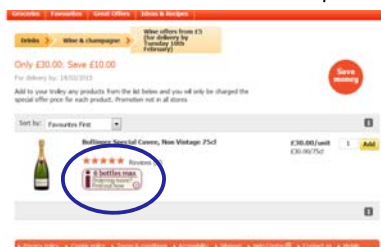
Explanations for anchoring effects

- Insufficient adjustment
- Priming (compatible associations)

Heuristics and Biases

Confirmation heuristic - Bias: Anchoring and Adjustment

Supermarket offers with a maximum limit per customer



Heuristics and Biases

More biases associated with confirmation heuristics:

- Conjunctive- and disjunctive event bias

Conjunctive events: events that must occur together with one another

Example: project planning

Project is finished on time only if all three steps are completed

Step 1: 90% finished on time

Step 2: 90% finished on time

Step 3: 90% finished on time

Probability that project is finished on time: $.9 \times .9 \times .9 = .73$

Problem: Typically people overestimate the probability of conjunctive events

Heuristics and Biases

Other biases associated with confirmation heuristics:

- Conjunctive- and disjunctive event bias

Disjunctive Events: A system fails if at least one bad event occurs

Example: Machine only works if all three parts work

Part1: 10% chance of failure

Part 2: 10% chance of failure

Part 3: 10% chance of failure

Probability that machine works: $.9 \times .9 \times .9 = .73$

Probability that machine fails: $1 - p(\text{machine works}) = 1 - .73 = .27$

Problem: Typically people underestimate the probability of disjunctive events

Heuristics and Biases

Other biases associated with confirmation heuristics:

- **Overconfidence**
Peoples subjective confidence in their judgments is greater than their objective accuracy
- **Hindsight bias**
Ex post recollections of ex ante guesses tilted in the direction of actual events ("I knew it!")
=> People think events are more predictable than they are
- **The curse of knowledge**
Difficult to imagine others know less (Examples: teachers, writers of computer manuals)

Heuristics and Biases

Summary

Heuristics:

- simplify cognitive complexity
- speed up decision making process
- often used inappropriately: cognitive shortcuts become shortcomings